

Project Documentation: Marketing Campaign & Customer Conversion Analytics

Executive BI Portfolio Project Specification

Problem Statement

Modern digital marketing campaigns generate high volumes of disparate data across ad spend, lead generation, and customer conversions. Marketers and executives often struggle to aggregate campaign metrics manually, leading to delayed decision-making, unoptimized budget allocation, and poor return on ad spend (ROAS). The objective of this project is to build an end-to-end data pipeline and an interactive two-page executive Power BI dashboard that monitors key financial and conversion metrics across marketing channels.

Objectives

- Design and deploy a robust Star Schema (Fact and Dimension tables) staged in MySQL.
- Clean and transform raw flat-file data using Power Query.
- Develop advanced DAX measures for executive KPIs (Total Leads, Spend, Revenue, and Conversion Rate %).
- Build a multi-page interactive executive Power BI dashboard with dynamic filtering slicers.
- Deliver clear visual insights into channel efficiency, spend vs. revenue, and acquisition share.

Tools & Architecture Used

- **Database Staging:** MySQL (Star Schema architecture with *lead_fact* and *campaign_dimension*)
- **Data Transformation:** Power Query (Header promotion, type casting, schema structuring)
- **Data Modeling & Calculation:** DAX (Data Analysis Expressions)
- **Visualization & BI:** Power BI Desktop (Multi-page report design)

Key Performance Indicators (KPIs) & DAX Measures

- **Total Leads:** $\text{Total Leads} = \text{COUNT}(\text{lead_fact}[\text{lead_id}])$
- **Total Ad Spend:** $\text{Total Spend} = \text{SUM}(\text{lead_fact}[\text{ad_spend}])$
- **Total Revenue:** $\text{Total Revenue} = \text{SUM}(\text{lead_fact}[\text{conversion_revenue}])$
- **Conversion Rate %:** $\text{Conversion Rate} = \text{DIVIDE}(\text{SUM}(\text{lead_fact}[\text{converted}]), \text{COUNT}(\text{lead_fact}[\text{lead_id}]), 0) * 100$

Visualizations Created (Multi-Page Report)

- **Page 1 (Executive Summary):** Top header banner, 4 executive KPI cards (Leads, Spend, Revenue, Conversion Rate), Clustered Column Chart for *Revenue by Marketing Channel*, Clustered Column Chart for *Ad Spend by Marketing Channel*, and an interactive Channel Slicer (List style).
- **Page 2 (Channel Distribution):** Donut chart for *Revenue Share by Marketing Channel*, Donut chart for *Ad Spend Share by Marketing Channel*, and a Horizontal Bar Chart for *Lead Volume by Channel*.

Expected Outcome

The executive dashboard provides an immediate, high-level snapshot of marketing ROI alongside granular channel contribution analysis. It empowers stakeholders to identify high-performing acquisition channels, eliminate inefficient ad spend, and optimize future budget allocation.